

|                                                                                                                                                                                                                                                                                                                                                                                                          |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <p>2. (a) (i) <math>V_0</math> = total volume of air required at <math>P_0</math>, including the residual air in the ball</p> $P_1 V_1 = P_0 V_0$ $(156 \text{ kPa}) (6000 \text{ cm}^3) = (100 \text{ kPa}) (V_0)$ $V_0 = 9360 \text{ cm}^3$ <p><math>\therefore</math> volume of air = <math>V_0</math> - volume of the basketball</p> $= 9360 \text{ cm}^3 - 6000 \text{ cm}^3$ $= 3360 \text{ cm}^3$ | 1M      |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 1A      |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 1M      |
| <div> <p>Alternatively:</p> <math display="block">\therefore n = \frac{PV}{RT}</math> <math display="block">\frac{156V}{RT} - \frac{100V}{RT} = n'</math> <math display="block">\therefore V' = \frac{56V}{RT} \cdot \frac{RT}{100} = 3360 \text{ cm}^3</math> </div>                                                                                                                                    | 1M      |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 1M + 1A |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 3       |
| <p>(ii) Number of strokes required = <math>3360 \text{ cm}^3 \div 120 \text{ cm}^3</math></p> $= 28$                                                                                                                                                                                                                                                                                                     | 1A      |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 1       |
| <p>(b) Using kinetic theory, the pressure <math>p</math> is given by</p>                                                                                                                                                                                                                                                                                                                                 |         |
| $pV = \frac{Nm\overline{c^2}}{3} = \frac{2N}{3} \cdot \frac{mc^2}{2}$                                                                                                                                                                                                                                                                                                                                    |         |
| <p>As the volume and the temperature (<math>\propto</math> kinetic energy of the air molecules) remain unchanged, the increase in pressure is due to the increase of number of air molecules hitting the wall of the container per unit time.</p>                                                                                                                                                        | 1A      |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 1A      |
| <p>Or Frequency of collision with the wall of container increases.</p>                                                                                                                                                                                                                                                                                                                                   | 1A      |
|                                                                                                                                                                                                                                                                                                                                                                                                          | 2       |

| Solution                                                                                                                                                                 |                                                                                                                                                                                                                                           | Marks                                                                                                                                                                      | Remarks |                                            |                        |          |                   |       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------------------------------|------------------------|----------|-------------------|-------|
| 5. (a)                                                                                                                                                                   | (i) $m (5.0 \text{ cm}) = 50 \text{ g} (10.0 \text{ cm})$<br>$m = 100 \text{ g or } 0.1 \text{ kg}$                                                                                                                                       | 1M<br>1A                                                                                                                                                                   |         |                                            |                        |          |                   |       |
|                                                                                                                                                                          |                                                                                                                                                                                                                                           | 2                                                                                                                                                                          |         |                                            |                        |          |                   |       |
|                                                                                                                                                                          | (ii) Counter-weight position : $10.0 \text{ cm} \pm 0.1 \text{ cm}$<br>Percentage error = $100\% \times \left(\frac{0.1}{10.0}\right) = 1\%$<br>$\therefore m = 101 \text{ g to } 99 \text{ g}$<br>i.e. maximum error = $\pm 1 \text{ g}$ | 1M<br>1A                                                                                                                                                                   |         |                                            |                        |          |                   |       |
|                                                                                                                                                                          |                                                                                                                                                                                                                                           | 2                                                                                                                                                                          |         |                                            |                        |          |                   |       |
|                                                                                                                                                                          | (b) Spring balance reading = $mg = (0.1 \text{ kg}) (9.81 \text{ N kg}^{-1})$<br>$= 0.981 \text{ N} (1.0 \text{ N for } g = 10 \text{ N kg}^{-1})$<br>$\approx 1.0 \text{ N}$                                                             | 1M/1A                                                                                                                                                                      |         |                                            |                        |          |                   |       |
|                                                                                                                                                                          |                                                                                                                                                                                                                                           | 1                                                                                                                                                                          |         |                                            |                        |          |                   |       |
|                                                                                                                                                                          | (c) (i)                                                                                                                                                                                                                                   | <table border="1"><tr><td>counter-weight position<br/>on beam balance</td><td>spring balance reading</td></tr><tr><td>the same</td><td>reading increases</td></tr></table> |         | counter-weight position<br>on beam balance | spring balance reading | the same | reading increases | 1A+1A |
|                                                                                                                                                                          | counter-weight position<br>on beam balance                                                                                                                                                                                                | spring balance reading                                                                                                                                                     |         |                                            |                        |          |                   |       |
|                                                                                                                                                                          | the same                                                                                                                                                                                                                                  | reading increases                                                                                                                                                          |         |                                            |                        |          |                   |       |
|                                                                                                                                                                          |                                                                                                                                                                                                                                           | 2                                                                                                                                                                          |         |                                            |                        |          |                   |       |
| (ii) The beam balance would fail to work / to measure the mass of the load,<br>as the apparent weight is zero (or weightless), the counter-weight can take any position. | 1A<br>1A                                                                                                                                                                                                                                  |                                                                                                                                                                            |         |                                            |                        |          |                   |       |
|                                                                                                                                                                          | 2                                                                                                                                                                                                                                         |                                                                                                                                                                            |         |                                            |                        |          |                   |       |